

Digital Signal Analysis and Processing Exam Notes

CT 704 (BEI/BCT, IV-I) / EX 753 (BEX, IV-II) · TU, IOE · tiers over 45 papers, 2066–2082 BS

6. IIR Filter Design

6 hrs · 50 questions across the 45 papers · worth about 13 of 80 marks · 4 syllabus sub-topics · 2nd of the 8 chapters by weight, behind only the DFT.

Read this first:

- **36 of the 50 questions are one calculation:** design an analog Butterworth low-pass filter, then map it to digital. Learn that one procedure and the chapter is mostly done.
- The whole subject of IIR design is **two sentences**. There is no good direct method in the z domain, but analog filter design is a solved, closed-form art. So **design in s , then map $s \rightarrow z$** .
- **Only two maps exist** in this syllabus, and they split the chapter:
 - **Bilinear** **TOP 36/45**: $s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$. One to one, so **no aliasing**, but the frequency axis is **warped**, so the edges must be pre-warped first.
 - **Impulse invariance** **TOP 11/45**: $h[n] = h_a(nT)$. The frequency axis is **linear**, so no warping, but sampling an unbandlimited analog response **aliases**.
- **Everything else is the analog half:** which prototype (Butterworth or Chebyshev), what order, what cut-off. That part is identical for both maps.
- **The sampling period cancels.** When the spec is given in rad/sample, pre-warp by $\Omega = \frac{2}{T} \tan \frac{\omega}{2}$ and map back by $s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$, and **every $2/T$ divides out**: $H(z)$ is the same for $T = 1$, $T = 2$ or $T = 0.1$. Nine papers give a sampling frequency purely as a distraction. **Set $T = 1$ and write one line saying why** (§6.5).
- The other 14 questions are bookwork: why transform at all, the two methods compared, pre-warping, and spectral transformation. All short, all below.

6.1 Why design through an analog prototype

Why do we use transformation of continuous time filter design methods to design discrete time IIR filters? **3+6 3**

PIN 2/45 70 Ma **3+6**, 73 Shr **3** · three reasons, one line each, then move on

The three reasons

- 1. The analog art is mature.** Continuous-time IIR design is a century old and its results are tabulated. Reusing it is free.
- 2. Closed-form formulas exist.** Butterworth, Chebyshev and elliptic prototypes all have closed-form order, cut-off and pole expressions, so the design is arithmetic, not search.
- 3. No such formulas exist in the z domain.** The same approximation methods applied directly to a discrete-time filter do *not* give closed forms, because a digital frequency response is **periodic** and an analog one is not.

The three steps that follow from it

- 1. Translate the spec** from digital ω to analog Ω , by the rule that belongs to the map being used.
- 2. Design $H_a(s)$:** choose the prototype family, find N and Ω_c , place the poles, write $H_a(s)$.
- 3. Map back** to $H(z)$ with the same rule.

What the map must preserve

- The $j\Omega$ axis of the s plane must land on the **unit circle** of the z plane, or the frequency response is not preserved.
- The **left half plane** must land **inside** the unit circle, or a stable analog filter becomes an unstable digital one.
- **Both maps below satisfy both conditions.** That is the whole reason there are only two of them.

6.2 The analog prototypes

Compare the Butterworth approximation with the Chebyshev approximation 1+10 3

PIN 2/45 76 Bh 1+10, 82 Bh · a table, not prose

Butterworth: maximally flat

$$|H_a(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

- **All pole.** Every zero is at $\Omega \rightarrow \infty$.
- $|H_a(j0)| = 1$ for every N , and $|H_a(j\Omega_c)| = \frac{1}{\sqrt{2}}$, i.e. Ω_c is the 3 dB point whatever the order.
- **Monotonic** in both bands: no ripple anywhere.
- Poles lie on a **circle** of radius Ω_c , equally spaced, none on the $j\Omega$ axis.

Chebyshev type I: equiripple passband

$$|H_a(j\Omega)|^2 = \frac{1}{1 + \varepsilon^2 V_N^2(\Omega/\Omega_c)}$$

- $V_N(x) = \cos(N \cos^{-1} x)$ for $|x| \leq 1$, which oscillates between ± 1 , and $\cosh(N \cosh^{-1} x)$ for $x > 1$, which climbs fast.
- So the passband **ripples** between 1 and $1/\sqrt{1 + \varepsilon^2}$, and the stopband falls monotonically and **steeply**.
- **Type II** puts the ripple in the stopband instead, by using V_N^{-1} ; **elliptic** ripples in both and is steeper still.
- $|H_a(j0)| = 1$ for N odd but $1/\sqrt{1 + \varepsilon^2}$ for N even. An even-order Chebyshev filter does **not** have unity DC gain.

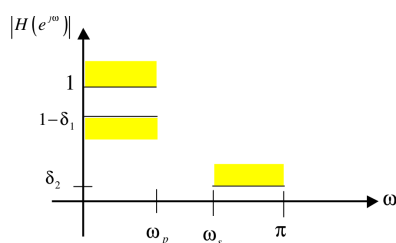
The comparison, as the answer

	Butterworth	Chebyshev I
Passband	monotonic, maximally flat	equiripple between 1 and $1/\sqrt{1 + \varepsilon^2}$
Stopband	monotonic	monotonic
Transition	widest of the three families	narrower for the same order
Order for a given spec	highest	lower, typically by a third
Poles lie on	a circle, radius Ω_c	an ellipse , semi-axes $a\Omega_c$ and $b\Omega_c$
Phase	most nearly linear of the three	worse, and worst near the band edge
Use when	flatness matters more than sharpness	a short filter matters more than flatness

The number to quote comes from the 45 papers' own recurring spec ($\omega_p = 0.2\pi$, $\omega_s = 0.3\pi$, 1 dB, 15 dB), asked by 73 Shr, 77 Ch and 76 Bh: Butterworth needs $N = 6$, Chebyshev only $N = 4$. That single sentence answers "compare" better than a paragraph. verified/added The local notes offer a second example, $\Omega_p = 1$, $\Omega_s = 3.33$, $\alpha_p = 0.3$ dB, $\alpha_s = 22$ dB, and print " $n = 5$ for butterworth, $n = 3$ for chebyshev". The Butterworth figure is wrong: the ratio is $157.489/0.071519 = 2202.05$, its log is 3.3429, and $2 \log(3.33) = 1.0449$, so $N = 3.1992$ and the answer is 4, not 5. The Chebyshev 3 is right. Use the paper-derived example above and this one only with the correction.

The design formulas, in the form every paper's specification uses core

these four lines carry 36 questions · memorise them exactly



Digital filter specifications

The tolerance scheme that every paper states in words. δ_1 is the passband deviation, δ_2 the stopband gain, and only the magnitude is ever specified: an IIR filter's phase is whatever the poles give.

Attenuation is always positive dB

$$\alpha = -20 \log_{10} |H|$$

- "maximum deviation 1 dB below 0 dB gain" $\Rightarrow \alpha_p = 1$.
- "maximum gain -15 dB in the stopband" $\Rightarrow \alpha_s = 15$.
- a gain 0.89125 given directly $\Rightarrow \alpha_p = -20 \log_{10} 0.89125 = 1$ dB.
- a ripple $\delta_p = \delta_1 = 0.11 \Rightarrow$ the edge gain is $1 - \delta_p = 0.89$, so $\alpha_p = 1.0122$ dB; but $\delta_s = \delta_2 = 0.21$ is a gain already, so $\alpha_s = 13.5556$ dB.
- Those four phrasings are the same spec four ways. Converting first, before anything else, is what makes the rest mechanical.

Order and cut-off

$$N \geq \frac{\log_{10} \left[\frac{10^{0.1\alpha_s} - 1}{10^{0.1\alpha_p} - 1} \right]}{2 \log_{10}(\Omega_s/\Omega_p)} \quad (\text{Butterworth})$$

$$N \geq \frac{\cosh^{-1} \sqrt{\frac{10^{0.1\alpha_s} - 1}{10^{0.1\alpha_p} - 1}}}{\cosh^{-1}(\Omega_s/\Omega_p)} \quad (\text{Chebyshev})$$

$$\Omega_c = \frac{\Omega_p}{(10^{0.1\alpha_p} - 1)^{1/2N}}, \quad \varepsilon = \sqrt{10^{0.1\alpha_p} - 1}$$

- **Always round N UP.** The fractional N meets the spec exactly and is not realisable.
- That Ω_c makes the **passband** exact and leaves the stopband over-satisfied. Using Ω_s and α_s instead makes the stopband exact. Both are legal; **state which you used.**
- Chebyshev needs no separate Ω_c : the passband is equiripple right up to Ω_p , so $\Omega_c = \Omega_p$.

Chebyshev poles

$$\alpha = \frac{1}{\varepsilon} + \sqrt{1 + \frac{1}{\varepsilon^2}}, \quad a = \frac{\alpha^{1/N} - \alpha^{-1/N}}{2}, \quad b = \frac{\alpha^{1/N} + \alpha^{-1/N}}{2}$$

$$s_k = -a\Omega_c \sin \theta_k + j b\Omega_c \cos \theta_k, \quad \theta_k = \frac{(2k+1)\pi}{2N}$$

- The poles sit on the **ellipse** $\frac{\sigma^2}{a^2\Omega_c^2} + \frac{\Omega^2}{b^2\Omega_c^2} = 1$, at the same angles a Butterworth filter would use, squashed towards the $j\Omega$ axis.
- $b > 1 > a$ always, so the ellipse is **taller than wide**: the poles crowd the imaginary axis, which is what sharpens the cut-off.

Butterworth poles

$$s_k = \Omega_c e^{j\pi \frac{N+2k+1}{2N}}, \quad k = 0, 1, \dots, N-1$$

- These are the N left-half-plane roots of $1 + (s/j\Omega_c)^{2N} = 0$; the other N belong to $H_a(-s)$ and are discarded.
- Equally spaced by π/N on the circle of radius Ω_c , symmetric about the real axis, **never on the $j\Omega$ axis.**
- A real pole at $-\Omega_c$ appears **only for odd N .**

$$H_a(s) = \frac{\Omega_c^N}{\prod_k (s - s_k)}$$

- The numerator Ω_c^N is fixed by $H_a(0) = 1$, not chosen.

Chebyshev gain, and the even-order trap

- $H_a(s) = \frac{K}{\prod_k (s - s_k)}$ with $K = \prod_k (-s_k)$ for **odd N** , which makes $H_a(0) = 1$.
- For **even N** the equiripple passband starts at its *lower* edge, so $H_a(0) = \frac{1}{\sqrt{1 + \varepsilon^2}}$ and **K is divided by $\sqrt{1 + \varepsilon^2}$.**
- Worked instance: 72 Ma has $\varepsilon = 1$, so its $N = 2$ filter has a DC gain of $1/\sqrt{2} = 0.707$, **not 1**. It looks like an arithmetic error and is not.
- ε and Ω_c come entirely from the passband. **Only N uses the stopband at all.**

6.3 The impulse invariance method

Describe digital Butterworth filter design using the impulse invariance technique. What are the limitations of the impulse invariance technique? **15 12+3**

TOP 11/45 69 Bh **15**, 78 Bh **12+3**, 72 Ch **3+12**, 81 Ch, 80 Ch, 69 Bh **12**, 70 Asa **13**, 74 Ch **12**, 67 Mng **15** · the limitations are half the marks

In one line: **make the digital impulse response the sampled analog one**, $h[n] = h_a(nT)$. Everything below follows from that single sentence, and so does everything wrong with the method.

The idea, and the whole derivation

- **Sample the analog impulse response:** $h[n] = h_a(nT)$. Nothing else is assumed.
- Expand $H_a(s)$ in partial fractions over its N simple poles p_k :

$$H_a(s) = \sum_{k=1}^N \frac{A_k}{s - p_k} \Rightarrow h_a(t) = \sum_k A_k e^{p_k t} u(t)$$

- Sample it and take the z transform, which is just a geometric series in $e^{p_k T} z^{-1}$:

$$H(z) = \sum_{n=0}^{\infty} \left(\sum_k A_k e^{p_k nT} \right) z^{-n} = \boxed{\sum_{k=1}^N \frac{A_k}{1 - e^{p_k T} z^{-1}}}$$

- So the rule is **term by term**: $\frac{1}{s - p_k} \rightarrow \frac{1}{1 - e^{p_k T} z^{-1}}$. That is the only thing to remember.
- **Spec translation is linear**: $\Omega = \omega/T$. **No pre-warping**.
- **The T convention**. Some texts set $h[n] = T h_a(nT)$ instead, which multiplies every A_k by T . Without it the filter's gain comes out near $1/T$, so with $T \neq 1$ you must say which you used. **With $T = 1$ s the two are identical**, and 8 of these 11 papers set $T = 1$.

The limitations (say all four)

1. **Aliasing**. Sampling gives $H(e^{j\omega}) = \frac{1}{T} \sum_k H_a(j\frac{\omega+2\pi k}{T})$. No practical analog prototype is band-limited, so the tails fold back and the response is **never exactly the analog one**.
2. **Low pass and band pass only**. A high-pass or band-stop prototype has an infinite high-frequency tail, so it aliases catastrophically. **Impulse invariance cannot make a high-pass filter**.
3. **Needs a partial fraction expansion** of $H_a(s)$, and it assumes the poles are **simple**. Repeated poles need a separate rule.
4. **Gain depends on T** , so the result must be rescaled.

What the map does to the planes

- Each pole moves as $p_k \rightarrow e^{p_k T}$, so $|z| = e^{\sigma_k T}$ and $\angle z = \Omega_k T$.
- $\sigma_k < 0 \Rightarrow |z| < 1$: **a stable analog filter always gives a stable digital one**.
- But $z = e^{sT}$ is **many to one**: every horizontal strip of width $2\pi/T$ in the s plane maps onto the whole z plane.
- **This applies to the POLES only, not to H** . $H(z) \neq H_a(s)|_s$ under any substitution; only the pole positions transform that simply.

The spec translation, which is all of step 1

- Take the tolerance scheme of §6.2 and **divide every frequency edge by T** . Nothing else changes: the bands $1 - \delta_1$ and δ_2 are untouched.
- The prototype is then specified only up to π/T , above which it is *assumed* negligible. **That assumption is exactly what aliasing violates**, which is why the design is checked at the end rather than trusted.
- Contrast §6.4, where the axis is **bent** rather than merely rescaled.
- Practical consequence: **over-design the stopband**. Extra attenuation there is what buys back the margin that the folded tails eat.

6.4 The bilinear transformation

Explain briefly the bilinear transformation method of IIR filter design, with its advantages and disadvantages **3 3+6 2+10**

TOP 36/45 73 Shr **3**, 70 Ma **3+6**, 81 Ba **2+10**, and the 32 design papers in §6.5

Where the formula comes from

- Take the simplest analog filter, $H_a(s) = \frac{b}{s+a}$, i.e.

$$\frac{dy}{dt} + ay(t) = bx(t).$$
- Integrate over one sample interval and apply the **trapezoidal rule**:

$$y(nT) - y(nT - T) = \frac{T}{2} [g(nT) + g(nT - T)], \quad g = bx - ay$$

- Take z transforms and solve. The result is $H_a(s)$ with

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$$

- So **the bilinear transformation IS trapezoidal integration**, which is why it is stable and one to one. That sentence is worth stating: it is the only derivation in the chapter and papers ask for it.
- Inverting, $z = \frac{1 + sT/2}{1 - sT/2}$.

What it does to the planes

- Put $s = \sigma + j\Omega$. Then

$$|z|^2 = \frac{(1 + \sigma T/2)^2 + (\Omega T/2)^2}{(1 - \sigma T/2)^2 + (\Omega T/2)^2}$$

- $\sigma < 0 \Rightarrow |z| < 1$, $\sigma = 0 \Rightarrow |z| = 1$, $\sigma > 0 \Rightarrow |z| > 1$.

The entire left half plane maps into the unit disc, exactly once.

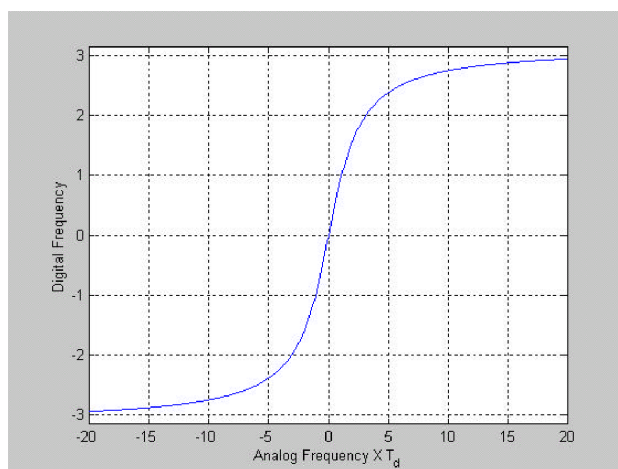
- So stability is preserved, and **there is no aliasing**: the map is one to one, not many to one.
- On the axis, $s = j\Omega$ gives $z = e^{j\omega}$ with

$$\omega = 2 \tan^{-1} \frac{\Omega T}{2},$$

$$\Omega = \frac{2}{T} \tan \frac{\omega}{2}$$

What is frequency warping, what is pre-warping, and why is it necessary **3+12 3**

HOT 4/45 78 Bh **3**, 82 Bh **3+12**, 72 Ch **3**, 79 Ba **3** · the figure below is the answer



Digital ω against analog ΩT_d . The curve is $\omega = 2 \tan^{-1}(\Omega T_d/2)$: a straight line through the origin at low frequency, flattening to the horizontal asymptotes $\pm\pi$.

Warping

- Ω runs over all of $(-\infty, \infty)$ but ω only over $(-\pi, \pi)$, so **the infinite analog axis is compressed into a finite digital one**. That compression is the warping.
- It is **almost linear for small ω** ($\tan x \approx x$) and **severely non-linear near $\omega = \pi$** .
- Consequence: a shape defined by more than its band edges, such as a differentiator or an arbitrary magnitude curve, comes out **distorted**. Bilinear design is only safe for **piecewise-constant** magnitude responses: low pass, high pass, band pass, band stop.

Pre-warping

- Because the map bends the axis, designing the analog filter at the *stated* edges puts the digital edges somewhere else.
- So **push the edges through the same bend first**:
 $\Omega_p = \frac{2}{T} \tan \frac{\omega_p}{2}$, $\Omega_s = \frac{2}{T} \tan \frac{\omega_s}{2}$.
- The warp then undoes the pre-warp exactly, and **the critical frequencies land where the specification asked**. Everything between them is still bent, which is why only the edges can be guaranteed.
- Skipping the pre-warp is the single most common lost mark in the chapter.**

6.5 Impulse invariance against bilinear

Compare the impulse invariance method with the bilinear transformation method 12+3 11+3 3+12 2+10

HOT 6/45 81 Bh 12+3, 79 Ba 11+3, 82 Ba 12+3, 72 Ch 3+12, 81 Ba 2+10, 70 Ma 3+6 · asked as a rider on a design question six times

	Impulse invariance	Bilinear
Basis	$h[n] = h_a(nT)$, a time-domain match	trapezoidal integration, a frequency-domain map
$s \rightarrow z$	poles only: $p_k \rightarrow e^{p_k T}$	$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$ throughout
Frequency map	$\Omega = \omega/T$, linear	$\Omega = \frac{2}{T} \tan \frac{\omega}{2}$, non-linear
Mapping type	many to one	one to one
Aliasing	yes , and unavoidable	none
Warping	none, shape preserved	yes, needs pre-warping
Filter types	low pass, band pass only	all four
Non-constant shapes	can do a band-limited differentiator	cannot
Needs	partial fractions, simple poles	algebraic substitution only
Order needed	higher, to absorb the aliasing	lower

The one-sentence verdict

- **Bilinear trades shape fidelity for the absence of aliasing; impulse invariance does the reverse.**
 - Since almost every filter wanted is piecewise constant, where shape fidelity between the edges is worth nothing, **bilinear wins almost always**, and 36 of this chapter's 50 questions use it.
 - The exception is a filter whose *shape* matters across the band. Only impulse invariance can make a band-limited differentiator.
- Why T cancels in a bilinear design** verified/added
- Pre-warping gives $\Omega_p = \frac{2}{T} \tan \frac{\omega_p}{2}$, so Ω_s/Ω_p is **free of T** and so is N .
 - Ω_c then carries one factor $\frac{2}{T}$, so every pole does, so $H_a(s)$ is a function of $sT/2$ alone.
 - Substituting $s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$ cancels it. **$H(z)$ is identical for every T .**
 - So a paper's "sampling frequency 0.5 Hz" changes nothing in the answer. Use $T = 1$, and **say in one line that it cancels** to show it was not ignored.
 - **This is false for impulse invariance**, where $\Omega = \omega/T$ scales the poles but $e^{p_k T}$ does not cancel it.

6.6 Spectral transformation in the digital domain

Why is spectral transformation required, and what are its features and parameters for low pass to high pass conversion 2 4 11+4

HOT 6/45 70 Asa 2, 80 Ba 4, 70 Ch 5, 79 Bh, 71 Ch, 69 Ch 11+4

Why it is required

- Every design procedure above produces a **low pass** filter. High pass, band pass and band stop have to come from somewhere.
- The analog route (transform $H_a(s)$ into a high pass, then discretise) works for bilinear but **fails for impulse invariance**, because a high-pass analog filter is not band-limited and aliases.
- So do it the other way: **discretise the low pass first**, then transform **inside the z domain**. Aliasing is then impossible, because nothing is sampled.
- **For bilinear designs the two routes give the identical filter**; for impulse invariance only the digital route exists. That is the answer to "why required".

What the map must satisfy

- $p^{-1} = G(z^{-1})$ must be **rational** in z^{-1} .
- It must take the unit circle to the unit circle, and the inside to the inside, so that **stability survives**.
- The general form that does both is an **all-pass** function, $\prod_k \frac{z^{-1} - \alpha_k}{1 - \alpha_k^* z^{-1}}$ with every $|\alpha_k| < 1$.
- **N all-pass factors give an N -fold map**, which is why the two band cases below need z^{-2} and the two low-pass-like ones need only z^{-1} .

The four transformations (this is the table they want)

To	Substitute $z^{-1} \rightarrow$	Parameters
Low pass	$\frac{z^{-1} - \alpha}{1 - \alpha z^{-1}}$	$\alpha = \frac{\sin \frac{\omega_p - \omega'_p}{2}}{\sin \frac{\omega_p + \omega'_p}{2}}$
High pass	$-\frac{z^{-1} + \alpha}{1 + \alpha z^{-1}}$	$\alpha = -\frac{\cos \frac{\omega_p + \omega'_p}{2}}{\cos \frac{\omega_p - \omega'_p}{2}}$
Band pass	$-\frac{z^{-2} - a_1 z^{-1} + a_2}{a_2 z^{-2} - a_1 z^{-1} + 1}$	$a_1 = \frac{2\alpha k}{k+1},$ $a_2 = \frac{k-1}{k+1}$
Band stop	$\frac{z^{-2} - a_1 z^{-1} + a_2}{a_2 z^{-2} - a_1 z^{-1} + 1}$	$a_1 = \frac{2\alpha}{k+1},$ $a_2 = \frac{1-k}{1+k}$

For the two band cases $\alpha = \frac{\cos \frac{\omega_u + \omega_L}{2}}{\cos \frac{\omega_u - \omega_L}{2}}$, and $k = \cot \frac{\omega_u - \omega_L}{2} \tan \frac{\omega_p}{2}$ for band pass, $\tan \frac{\omega_u - \omega_L}{2} \tan \frac{\omega_p}{2}$ for band stop. ω_p is the prototype's edge, ω'_p the new one.

Reading the table without mistakes

- **Only the low-pass row keeps the sign**; the high-pass row carries a leading minus, and that minus is the whole reason the passband moves to $\omega = \pi$.
- The substitution goes into $H_{lp}(p)$ **written in p^{-1}** , not in p . Every published prototype already is.
- **Every zero at $z = -1$ becomes a zero at $z = +1$** , so a high pass built this way has $H(1) = 0$ exactly: it kills DC. **Free check on the answer.**
- $|\alpha| < 1$ always. If it comes out bigger, the two edge frequencies have been swapped in the cosine.
- Both band rows raise the order: an N th-order low pass becomes a **$2N$ th-order** band pass or band stop.

6.7 Last-minute recall

Say these without thinking

- $\alpha = -20 \log_{10} |H|$, and δ_p means an edge gain of $1 - \delta_p$.
- $N = \frac{\log [(10^{0.1\alpha_s} - 1)/(10^{0.1\alpha_p} - 1)]}{2 \log(\Omega_s/\Omega_p)}$, **round up**.
- $\Omega_c = \Omega_p / (10^{0.1\alpha_p} - 1)^{1/2N}$.
- Bilinear: pre-warp $\Omega = \frac{2}{T} \tan \frac{\omega}{2}$, map back $s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}$, and **T cancels**.
- Impulse invariance: $\Omega = \omega/T$, and $\frac{1}{s-p_k} \rightarrow \frac{1}{1-e^{p_k T} z^{-1}}$.
- Butterworth poles: radius Ω_c , angles $\pi \frac{N+2k+1}{2N}$, spacing π/N .
- Chebyshev: $\varepsilon = \sqrt{10^{0.1\alpha_p} - 1}$, $\Omega_c = \Omega_p$, poles on an ellipse.
- LP $\rightarrow$ HP: $z^{-1} \rightarrow -\frac{z^{-1} + \alpha}{1 + \alpha z^{-1}}$, $\alpha = -\frac{\cos \frac{\omega_p + \omega'_p}{2}}{\cos \frac{\omega_p - \omega'_p}{2}}$.
- $\omega = 2\pi f / f_s$ when the spec is in Hz.

Mistakes that cost marks

- **Forgetting to pre-warp** in a bilinear design, or pre-warping in an impulse-invariance one.
- Rounding N down, or to nearest.
- Taking α_s as negative because the paper prints -15 dB. The formula wants $+15$.
- Reading $\delta_p = 0.11$ as a gain of 0.11. It is a **deviation**: the gain is 0.89.
- Using all $2N$ roots of the Butterworth equation. **Only the left half**.
- Substituting $z = e^{sT}$ into $H_a(s)$ for impulse invariance. That is not the method; only the **poles** move that way.
- Giving an even-order Chebyshev filter a DC gain of 1. It is $1/\sqrt{1 + \varepsilon^2}$.
- Claiming impulse invariance can make a high-pass filter. **It cannot**.
- Carrying T through a bilinear design and getting a T -dependent answer. **If T survives, there is an arithmetic error**.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. → reads “leads to / therefore”, + reads “and”, = reads “is”. ✓ means the property holds, × that it does not.

Short	Means	Short	Means
DSP	digital signal processing	DSAP	digital signal analysis and processing
DT	discrete-time	CT	continuous-time
LTI	linear time-invariant	LSI	linear shift-invariant, the same thing
TI	time-invariant	BIBO	bounded input, bounded output
LCCDE	linear constant-coefficient difference equation	ROC	region of convergence
DTFS	discrete-time Fourier series	DTFT	discrete-time Fourier transform
CTFT	continuous-time Fourier transform	DFS	discrete Fourier series
DFT	discrete Fourier transform	IDFT	inverse DFT
FFT	fast Fourier transform	DIT	decimation in time
DIF	decimation in frequency	MAC	multiply and accumulate
FIR	finite impulse response	IIR	infinite impulse response
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
BLT	bilinear transformation	II	impulse invariance
A/D	analog to digital	D/A	digital to analog
S/H	sample and hold	ZOH	zero-order hold
SNR	signal to noise ratio	PYQ	past-year question
coeff	coefficient	freq	frequency
eqn	equation	no.	number
ops	operations	attr	attribute
w/	with	w/o	without
Eg	for example	Vs	compare / differentiate
+Fig	a figure is wanted	(I)	importance / need
(+)	advantage	(-)	disadvantage

Exam paper codes

Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months**: 2075 Ashwin and 2070 Ashad are separate papers.

How a year code is written

- **Bold** = **Regular** exam. Plain = Back exam.
 - “New Back (2066 & Later Batch)” counts as Back.
 - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II). Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 **Bh** is EX 753 Regular, 74 **Ma** is EX 753 Back.
- **69 Bh and 69 Bh are two different papers**, one in each course.

Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.